MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW-POWER SHCOTTKY TTL,

PRIORITY ENCODERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low-power Schottky, TTL, priority encoder, logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	8-line to 3-line priority encoder
02	8-line to 3-line priority encoder with 3-state outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
X	C-2 (20-terminal .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
Device type 01	110 mW
Device type 02	137.5 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Case E	50°C/W
Case F	70°C/W
Case X	60°C/W
Junction temperature (T_J)	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.7 V dc
Normalized fanout (each output) 2/	10 maximum
Case operating temperature range	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be in accordance with MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figures 1 and 2, respectively.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schemtaics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and shall apply over the full recommended case operating temperature range.

2/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$ 1/	Device type	Limits Min	Limits Max	Unit
High-level output level	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -.4 mA	All	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 4.0 mA	01		0.40	V
		V_{CC} = 4.5 V, I_{OL} = 4.0 mA, EO, GS outputs	02			
		I_{OL} = 12 mA A0, A1, A2 outputs				
Input clamp volatge	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = +25°C	All		-1.5	V
Low level input current at data inputs except 0	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	All	-.26	-.76	mA
Low level input current at enable input and 0 input	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	All	-.12	-.40	mA
High level input current at data inputs	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All		40	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	All		200	µA
High level input current at enable input	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All		20	µA
	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	All		100	µA
Off-state output current high level voltage applied	I_{OZH}	V_{CC} = 5.5 V, V_{IN} = 2.0 V, V_O = 2.7 V	02		20	µA
Off-state output current low level voltage applied	I_{OZL}	V_{CC} = 5.5 V, V_{IN} = 2.0 V, V_O = 0.4 V	02		-20	-µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{OUT} = GND 1/	01	-15	-100	mA
		EO, GS	02	-15	-100	mA
		A0-A2	02	-30	-130	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Supply current	I_{CC1}	V_{CC} = 5.5 V	01		20	mA
			02		25	mA
	I_{CC2}	V_{CC} = 5.5 V	01		17	mA
			02		23	mA
Propagation delay time low-to-high level output from data input to A. In-phase output	t_{PLH1}	V_{CC} = 5.0 V dc R_L = 2/ C_L = 50 pF ±10%	01	2	30	ns
			02	2	29	ns
Propagation delay time high-to-low level output from data input to A. In-phase output	t_{PHL1}		01	2	39	ns
			02	2	46	ns
Propagation delay time low-to-high level output from data to A. Out-of-phase	t_{PLH2}		01	2	53	ns
			02	2	52	ns
Propagation delay time high-to-low level output from data to A. Out of phase	t_{PHL2}		01	2	44	ns
			02	2	52	ns
Propagation delay time low-to-high level output from data to E0	t_{PLH3}		01	2	30	ns
			02	2	30	ns
Propagation delay time high-to-low level output from data to E0	t_{PHL3}		01	2	58	ns
			02	2	58	ns
Propagation delay time low-to-high level output from data to GS	t_{PLH4}		01	2	78	ns
			02	2	78	ns
Propagation delay time high-to-low level output from data to GS	t_{PHL4}		01	2	34	ns
			02	2	34	ns
Propagation delay time low-to-high level output from E1 to GS	t_{PLH6}		01	2	29	ns
			02	2	29	ns
Propagation delay time high-to-low level output from E1 to GS	t_{PHL6}		01	2	53	ns
			02	2	53	ns

See footnotes at end of table.

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tions the manufacturer shall supply product capable of passing any screening or TCI (QCI) of testing modifica

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time low-to-high level output from E1 to E0	t_{PLH7}	V_{CC} = 5.0 V dc R_L = 2/ C_L = 50 pF ±10%	01 02	2 2	34 34	ns ns
Propagation delay time high-to-low level output from E1 to E0	t_{PHL7}		01 02	2 2	52 52	ns ns
Propagation delay time low-to-high level output E1 to A	t_{PLH5}		01	2	39	ns
Propagation delay time high-to-low level output from E1 to A	t_{PHL5}		01	2	39	ns
Enable time to high level output from E1 to A	t_{PZH}		02	2	64	ns
Enable time to low level output from E1 to A	t_{PZL}		02	2	60	ns
Disable time from high level output from E1 to A	t_{PHZ}		02	2	42	ns
Disable time from low level output from E1 to A	t_{PLZ}		02	2	52	ns

1/ Not more than one output should be shorted at one time.

2/ R_L = 2 kΩ ±5% for type 01 and E0, GS outputs of type 02 (regular output).
R_L = 316Ω ±5% for A0, A1, A2 outputs of 02 (tri-state output).

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-M-38510, appendix E).

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9,10,11	1*,2,3, 7,9
Group A test requirements (method 5005)	1,2,3,7, 8,9,10,11	1,2,3,7, 9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7, 8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(a) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(b) T_A = +125°C minimum.

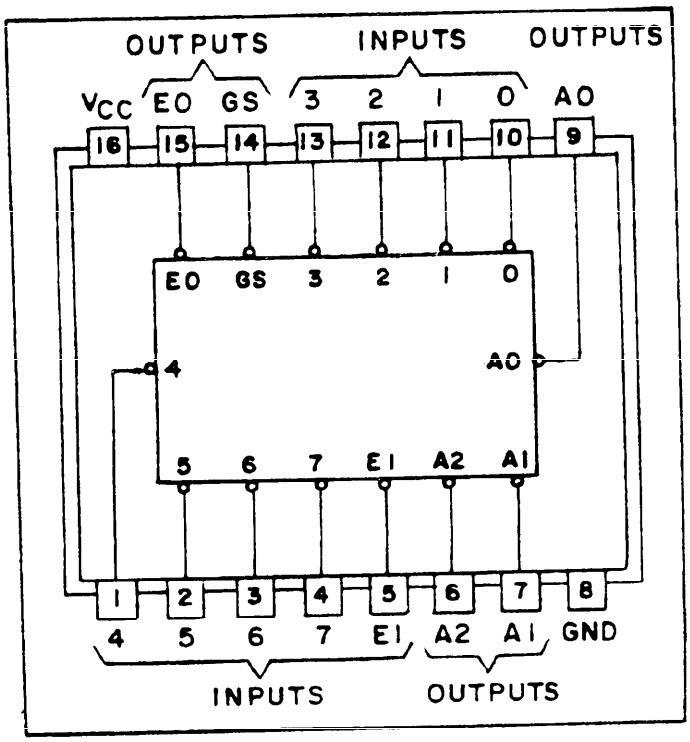
b. Interim and final electrical tests shall be as specified in table II, except interim electrical tests prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 tests after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

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Device types 01 and 02



Device types 01 and 02

Case X

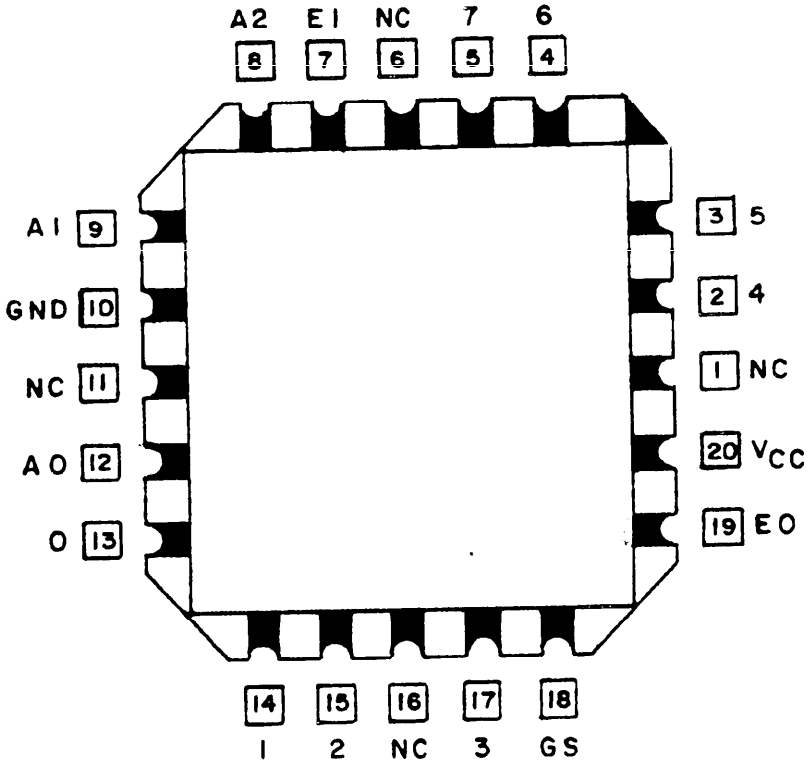


FIGURE 1. Terminal configurations.

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Device type 01

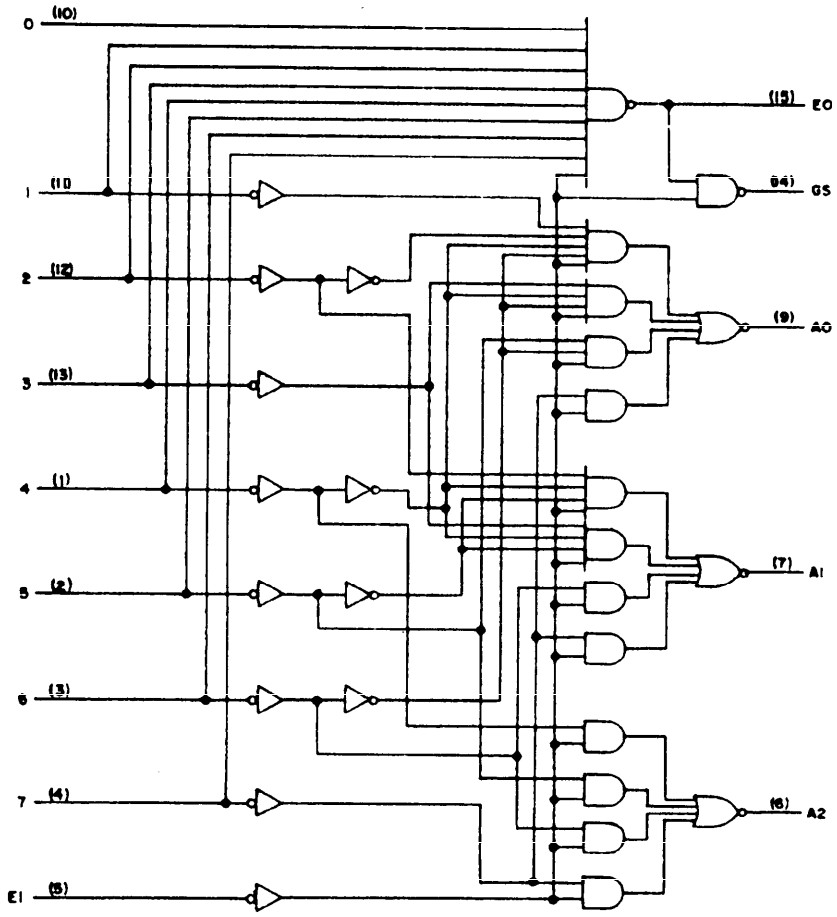


FIGURE 2. Logic diagrams.

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Device type 02

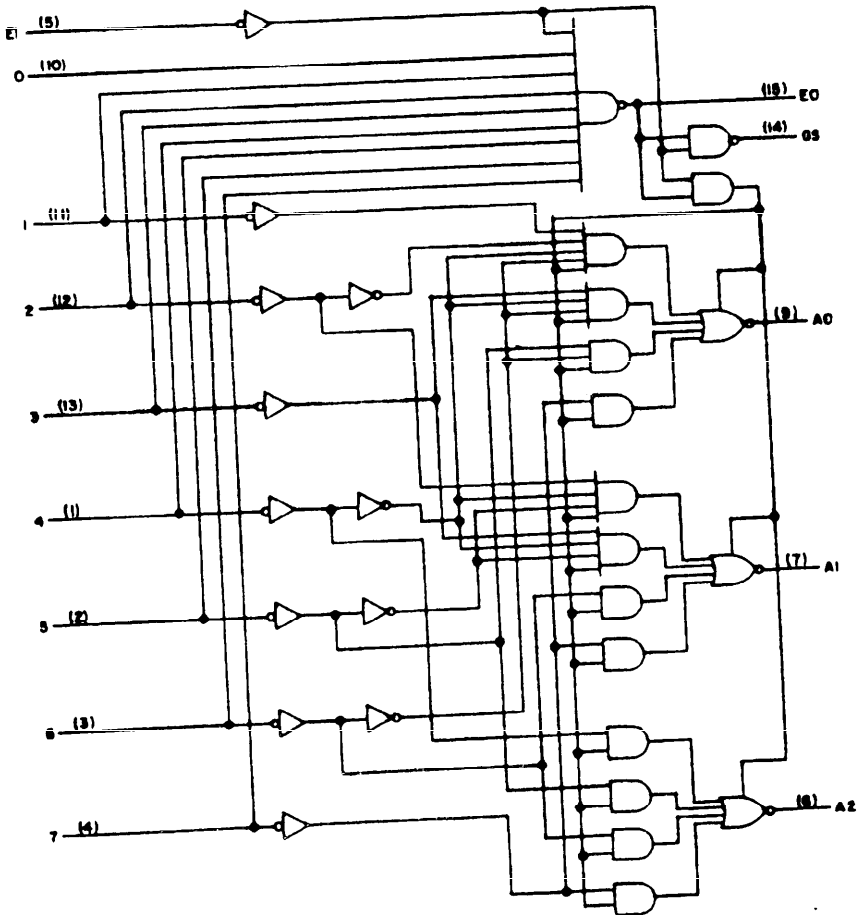


FIGURE 2. Logic diagrams - Continued.

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Device type 01

Function table

Inputs									Outputs				
E1	0	1	2	3	4	5	6	7	A2	A1	A0	GS	E0
H	X	X	X	X	X	X	X	X	H	H	H	H	H
L	H	H	H	H	H	H	H	H	H	H	H	H	L
L	X	X	X	X	X	X	X	L	L	L	L	L	H
L	X	X	X	X	X	X	L	H	L	L	H	L	H
L	X	X	X	X	X	L	H	H	L	H	L	L	H
L	X	X	X	X	L	H	H	H	L	H	H	L	H
L	X	X	X	L	H	H	H	H	H	L	L	L	H
L	X	X	L	H	H	H	H	H	H	L	H	L	H
L	X	L	H	H	H	H	H	H	H	H	L	L	H
L	L	H	H	H	H	H	H	H	H	H	H	L	H

Device type 02

Function table

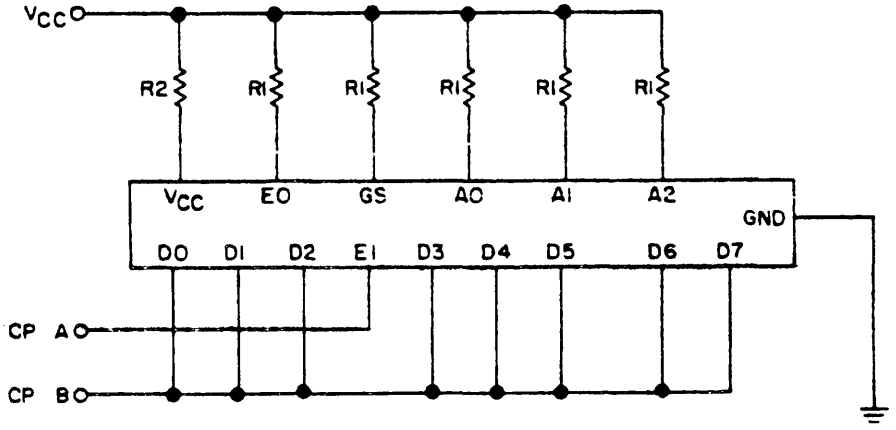
Inputs									Outputs				
E1	0	1	2	3	4	5	6	7	A2	A1	A0	GS	E0
H	X	X	X	X	X	X	X	X	Z	Z	Z	H	H
L	H	H	H	H	H	H	H	H	Z	Z	Z	H	L
L	X	X	X	X	X	X	X	L	L	L	L	L	H
L	X	X	X	X	X	X	L	H	L	L	H	L	H
L	X	X	X	X	X	L	H	H	L	H	L	L	H
L	X	X	X	X	L	H	H	H	L	H	H	L	H
L	X	X	X	L	H	H	H	H	H	L	L	L	H
L	X	X	L	H	H	H	H	H	H	L	H	L	H
L	X	L	H	H	H	H	H	H	H	H	L	L	H
L	L	H	H	H	H	H	H	H	H	H	H	L	H

H = High logic level, L = Low logic level, X = Irrelevant
Z = High-impedance state

FIGURE 3. Truth tables.

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Device types 01, 02

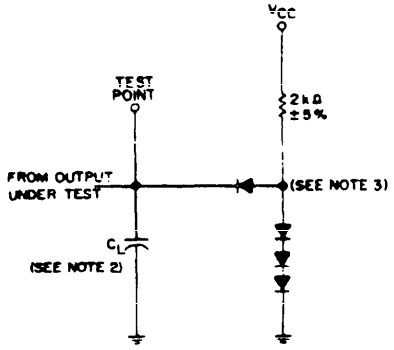


NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same as CPA, synchronized with CPA, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%; R2 = 10Ω max.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

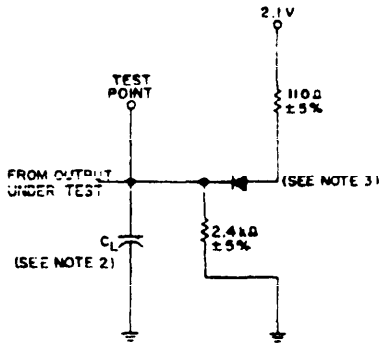
FIGURE 4. Burn-in and life test circuit.

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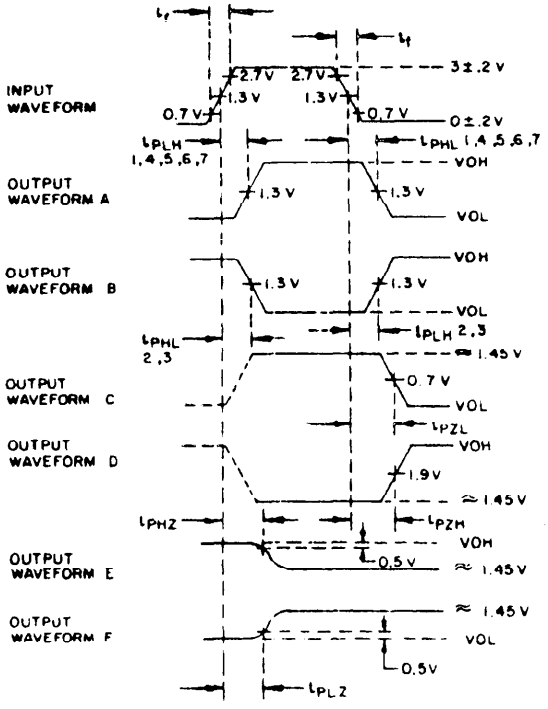
50 1.1 Carrier and container



LOAD CIRCUIT FOR DEVICE TYPE 01 (AND E_0, G_5 OUTPUTS FOR TYPE 02 - REGULAR OUTPUT)



LOAD CIRCUIT FOR A_0, A_1, A_2 OUTPUTS OF DEVICE TYPE 02 (TRI-STATE OUTPUT)



NOTES:
1. Input pulse characteristics: PRR ≤ 1.0 MHz, t_r ≤ 15 ns, and t_f ≤ 6 ns.
2. C_L = 50 pF ±10%.
   C_L includes probe and jig capacitance.
3. All diodes are 1N3064 or 1N916.
4. Load circuits on a given output are only required where the specific test given in table III indicates "OUT" on that output. Load circuits may otherwise be omitted.

FIGURE 5. Switching test.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Case X 1/ / Test no.	1 / 2 / 4	2 / 3 / 5	3 / 4 / 6	4 / 5 / 7	5 / 7 / E1	6 / 8 / A2	7 / 9 / A1	8 / 10 / GND	9 / 12 / A0	10 / 13 / 0	11 / 14 / 1	12 / 15 / 2	13 / 17 / 3	14 / 18 / GS	15 / 19 / E0	16 / 20 / V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1					2.0 V	-400 µA	-400 µA	GND	-400 µA					-400 µA	-400 µA	4.5 V	A2	2.5		V
	"	"	2					"	"	"	"	"							"	A1	"		"
	"	"	3					"	"	"	"	"							"	A0	"		"
	"	"	4					"	"	"	"	"							"	GS	"		"
	"	"	5					"	"	"	"	"							"	E0	"		"
	V_{OL}	3007	6				0.7 V	0.7 V	4 mA	4 mA	"	4 mA					4 mA	4 mA	"	A2		.4	"
	"	"	7				"	"	"	"	"	"					"	"	"	A1		"	"
	"	"	8				"	"	"	"	"	"					"	"	"	A0		"	"
	"	"	9				"	"	"	"	"	"					"	"	"	GS		"	"
	"	"	10	2.0 V	2.0 V	2.0 V	2.0 V	"	"	"	"	"	2.0 V	2.0 V	2.0 V	2.0 V	"	"	"	E0		"	"
	I_{IL1}	3009	11	0.4 V	5.5 V	5.5 V	5.5 V	GND			"		5.5 V	5.5 V	5.5 V	5.5 V			5.5 V	4	-.26	-.76	mA
	"	"	12	5.5 V	0.4 V	5.5 V	"	"			"		"	"	"	"			"	5	"	"	"
	"	"	13	"	5.5 V	0.4 V	"	"			"		"	"	"	"			"	6	"	"	"
	"	"	14	"	"	5.5 V	0.4 V	"			"		"	"	"	"			"	7	"	"	"
	I_{IL2}	"	15	"	"	"	5.5 V	"			"		0.4 V	"	"	"			"	0	-.16	-.40	"
	I_{IL1}	"	16	"	"	"	"	"			"		5.5 V	0.4 V	"	"			"	1	-.26	-.76	"
	"	"	17	"	"	"	"	"			"		"	5.5 V	0.4 V	"			"	2	"	"	"
	"	"	18	"	"	"	"	"			"		"	"	5.5 V	0.4 V			"	3	"	"	"
	I_{IL2}	"	19				"	0.4 V			"								"	E1	-.12	-.36	"
	I_{IH1}	3010	20	2.7 V							"								"	4		40	µA
	"	"	21		2.7 V						"								"	5		"	"
	"	"	22			2.7 V					"								"	6		"	"
	"	"	23				2.7 V				"								"	7		"	"
	"	"	24								"		2.7 V						"	0		"	"
	"	"	25								"			2.7 V					"	1		"	"
	"	"	26								"				2.7 V				"	2		"	"
	"	"	27								"					2.7 V			"	3		"	"
	I_{IH2}	"	28					2.7 V			"								"	E1		20	"
	I_{IH3}	"	29	5.5 V							"								"	4		700	"
	"	"	30		5.5 V						"								"	5		"	"
	"	"	31			5.5 V					"								"	6		"	"
	"	"	32				5.5 V				"								"	7		"	"
	"	"	33								"		5.5 V						"	0		"	"
	"	"	34								"			5.5 V					"	1		"	"
	"	"	35								"				5.5 V				"	2		"	"
	"	"	36								"					5.5 V			"	3		"	"
	I_{IH4}	"	37					5.5 V			"								"	E1		100	"
	I_{OS}	3011	38					4.5 V	GND	GND	"	GND					GND	GND	"	A2	-15	100	mA
	"	"	39					"	"	"	"	"					"	"	"	A1	"	"	"
	"	"	40					"	"	"	"	"					"	"	"	A0	"	"	"
	"	"	41					"	"	"	"	"					"	"	"	GS	"	"	"
	"	"	42					"	"	"	"	"					"	"	"	E0	"	"	"

See footnotes at end of device type 01.

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Table III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case X 1/ Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				4	5	6	7	E1	A2	A1	GND	A0	0	1	2	3	GS	E0	V_{CC}				
1 T_C = +25°C	I_{CC1}	3005	43				GND	GND			GND								5.5 V	V_{CC}		20	mA
	I_{CC2}	3005	44								"								"	V_{CC}		17	"
	V_{IC}		45	-18 mA							"								4.5 V	4		-1.5	V
			46		-18 mA						"								"	5		"	"
			47			-18 mA					"								"	6		"	"
			48				-18 mA				"								"	7		"	"
			49					-18 mA			"								"	E1		"	"
			50								"		-18 mA						"	0		"	"
			51								"			-18 mA					"	1		"	"
			52								"				-18 mA				"	2		"	"
			53								"					-18 mA			"	3		"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 2/ T_C = +25°C	Truth table tests	3014	54	A	A	A	A	A	H	H	GND	H	A	A	A	A	H	H	5.0 V	3/			
		"	55	"	"	"	A	B	H	H	"	H	"	"	"	"	H	L	"	"			
		"	56	"	"	B	B	A	L	L	"	L	"	"	"	"	L	H	"	"			
		"	57	"	B	A	"	"	"	L	"	H	"	"	"	"	"	L	"	"			
		"	58	"	A	"	"	"	"	H	"	L	"	"	"	"	"	"	"	"			
		"	59	B	"	"	"	"	"	H	"	L	"	"	"	"	"	"	"	"			
		"	60	A	"	"	"	"	H	L	"	H	"	"	"	B	"	"	"	"			
		"	61	"	"	"	"	"	"	L	"	L	"	"	B	A	"	"	"	"			
		"	62	"	"	"	"	"	"	H	"	H	"	B	A	"	"	"	"	"			
		"	63	"	"	"	"	"	"	"	"	L	B	A	"	"	"	"	"	"			
		"	64	"	"	"	"	A	"	"	"	H	A	"	"	"	H	"	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 Fig. 5	65	5.0 V	5.0 V	5.0 V	IN	GND		OUT	GND	OUT		5.0 V	5.0 V	5.0 V			5.0 V	7 to A0	2	23	ns
		"	66	"	"	"	"	"	OUT	"	"			"	"	"			"	7 to A1	"	"	"
		"	67	"	"	"	"	"	"	OUT	"			"	"	"			"	7 to A2	"	"	"
		"	68	"	"	IN	5.0 V	"	"	"	"			"	"	"			"	6 yo A1	"	"	"
		"	69	"	"	IN	"	"	"	OUT	"			"	"	"			"	6 to A2	"	"	"
		"	70	"	IN	5.0 V	"	"	"	"	"	OUT		"	"	"			"	5 to A0	"	"	"
		"	71	"	IN	"	"	"	"	OUT	"			"	"	"			"	5 to A2	"	"	"
		"	72	IN	5.0 V	"	"	"	"	OUT	"			"	"	"			"	4 to A2	"	"	"
		"	73	5.0 V	"	"	"	"	"	"	"	OUT		"	"	IN			"	3 to A0	"	"	"
		"	74	"	"	"	"	"	"	OUT	"			"	"	IN			"	3 to A1	"	"	"
		"	75	"	"	"	"	"	"	"	"			"	IN	5.0 V			"	2 to A1	"	"	"
		"	76	"	"	"	"	"	"	"	"	OUT		IN	5.0 V	"			"	1 to A0	"	"	"
	t_{PHL1}	"	77	"	"	"	"	"		OUT	"	OUT		5.0 V	"	"			"	7 to A0	"	30	"
		"	78	"	"	"	"	"	OUT	"	"			"	"	"			"	7 to A1	"	"	"
		"	79	"	"	"	"	"	"	OUT	"			"	"	"			"	7 to A2	"	"	"
		"	80	"	"	IN	5.0 V	"	"	"	"			"	"	"			"	6 yo A1	"	"	"
		"	81	"	"	IN	"	"	"	OUT	"			"	"	"			"	6 to A2	"	"	"
		"	82	"	IN	5.0 V	"	"	"	"	"	OUT		"	"	"			"	5 to A0	"	"	"
		"	83	"	IN	"	"	"	"	OUT	"			"	"	"			"	5 to A2	"	"	"
		"	84	IN	5.0 V	"	"	"	"	OUT	"			"	"	"			"	4 to A2	"	"	"
		"	85	5.0 V	"	"	"	"	"	"	"	OUT		"	"	IN			"	3 to A0	"	"	"
		"	86	"	"	"	"	"	"	OUT	"			"	"	IN			"	3 to A1	"	"	"
		"	87	"	"	"	"	"	"	OUT	"			"	IN	5.0 V			"	2 to A1	"	"	"
		"	88	"	"	"	"	"	"	"	"	OUT		IN	5.0 V	5.0 V			"	1 to A0	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X 1/ / Test no.	1 / 2 / 4	2 / 3 / 5	3 / 4 / 6	4 / 5 / 7	5 / 7 / E1	6 / 8 / A2	7 / 9 / A1	8 / 10 / GND	9 / 12 / A0	10 / 13 / 0	11 / 14 / 1	12 / 15 / 2	13 / 17 / 3	14 / 18 / GS	15 / 19 / EO	16 / 20 / V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
T_C = +25°C	t_{PLH2}	3003 Fig. 5	89	IN	5.0 V	5.0 V	5.0 V	GND		OUT	GND	OUT				GND			5.0 V	4 to A1	2	41	ns
			90	IN	"	"	"	"			"	OUT				GND			"	4 to A0	"	"	"
			91		GND	IN														6 to A0			
	t_{PHL2}	"	92	IN	5.0 V	5.0 V	"	"		OUT	"	OUT				GND			"	4 to A1	"	34	"
			93	IN	5.0 V	5.0 V	"	"			"	OUT				GND			"	4 to A0	"	"	"
			94		GND	IN														6 to A0			
	t_{PLH3}	"	95	5.0 V	5.0 V	5.0 V	"	"			"		IN	5.0 V	5.0 V	5.0 V		OUT	"	0 to EO	"	23	"
			96	"	"	"	"	"			"		5.0 V	IN	5.0 V	"		"	"	1 to EO	"	"	"
			97	"	"	"	"	"			"		"	5.0 V	IN	IN		"	"	2 to EO	"	"	"
			98	"	"	"	"	"			"		"	"	5.0 V	5.0 V		"	"	3 to EO	"	"	"
			99	IN	"	"	"	"			"		"	"	"	"		"	"	4 to EO	"	"	"
			100	5.0 V	IN	"	"	"			"		"	"	"	"		"	"	5 to EO	"	"	"
			101	"	5.0 V	IN	"	"			"		"	"	"	"		"	"	6 to EO	"	"	"
			102	"	"	5.0 V	IN	"			"		"	"	"	"		"	"	7 to EO	"	"	"
	t_{PHL3}	"	103	"	"	"	5.0 V	"			"		IN	"	"	"		"	"	0 to EO	"	45	"
			104	"	"	"	"	"			"		5.0 V	IN	"	"		"	"	1 to EO	"	"	"
			105	"	"	"	"	"			"		"	5.0 V	IN	IN		"	"	2 to EO	"	"	"
			106	"	"	"	"	"			"		"	"	5.0 V	5.0 V		"	"	3 to EO	"	"	"
			107	IN	"	"	"	"			"		"	"	"	"		"	"	4 to EO	"	"	"
			108	5.0 V	IN	"	"	"			"		"	"	"	"		"	"	5 to EO	"	"	"
			109	"	5.0 V	IN	"	"			"		"	"	"	"		"	"	6 to EO	"	"	"
			110	"	"	5.0 V	IN	"			"		"	"	"	"		"	"	7 to EO	"	"	"
	t_{PLH4}	"	111	5.0 V	5.0 V	5.0 V	"	"			"		IN	"	"	"	OUT	"	"	0 to GS	"	66	"
			112	"	"	"	"	"			"		5.0 V	IN	"	"	"	"	"	1 to GS	"	"	"
			113	"	"	"	"	"			"		"	5.0 V	IN	IN	"	"	"	2 to GS	"	"	"
			114	"	"	"	"	"			"		"	"	5.0 V	5.0 V	"	"	"	3 to GS	"	"	"
			115	IN	"	"	"	"			"		"	"	"	"	"	"	"	4 to GS	"	"	"
			116	5.0 V	IN	"	"	"			"		"	"	"	"	"	"	"	5 to GS	"	"	"
			117	"	5.0 V	IN	"	"			"		"	"	"	"	"	"	"	6 to GS	"	"	"
			118	"	"	5.0 V	IN	"			"		"	"	"	"	"	"	"	7 to GS	"	"	"
	t_{PHL4}	"	119	"	"	"	5.0 V	"			"		IN	"	"	"	"	"	"	0 to GS	"	26	"
			120	"	"	"	"	"			"		5.0 V	IN	"	"	"	"	"	1 to GS	"	"	"
			121	"	"	"	"	"			"		"	5.0 V	IN	IN	"	"	"	2 to GS	"	"	"
			122	"	"	"	"	"			"		"	"	5.0 V	5.0 V	"	"	"	3 to GS	"	"	"
			123	IN	"	"	"	"			"		"	"	"	"	"	"	"	4 to GS	"	"	"
			124	5.0 V	IN	"	"	"			"		"	"	"	"	"	"	"	5 to GS	"	"	"
			125	"	5.0 V	IN	"	"			"		"	"	"	"	"	"	"	6 to GS	"	"	"
			126	"	5.0 V	5.0 V	IN	"			"		"	"	"	"	"	"	"	7 to GS	"	"	"
	t_{PLH5}	"	127				GND	IN				OUT							"	E1 to A0	"	30	"
			128				"	"	OUT	OUT	"								"	E1 to A1	"	"	"
			129																	E1 to A2			
	t_{PHL5}	"	130				"	"			"	OUT							"	E1 to A0	"	"	"
			131				"	"	OUT	OUT	"								"	E1 to A1	"	"	"
			132																	E1 to A2			
	t_{PLH6}	"	133				"	"			"						OUT		"	E1 to GS	"	22	"

MIL-M-38510/10/5NDA

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TABLE I. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X 1/ / Test no.	1 / 2 / 4	2 / 3 / 5	3 / 4 / 6	4 / 5 / 7	5 / 7 / E1	6 / 8 / A2	7 / 9 / A1	8 / 10 / GND	9 / 12 / A0	10 / 13 / 0	11 / 14 / 1	12 / 15 / 2	13 / 17 / 3	14 / 18 / GS	15 / 19 / EO	16 / 20 / V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PHL6}	3003 Fig. 5	134				GND	IN			GND						OUT		5.0 V	E1 to GS	2	41	ns
	t_{PLH7}	"	135	5.0 V	5.0 V	5.0 V	5.0 V	"			"		5.0 V	5.0 V	5.0 V	5.0 V		OUT	"	E1 to EO	"	26	"
	t_{PHL7}	"	136	5.0 V	5.0 V	5.0 V	5.0 V	"			"		5.0 V	5.0 V	5.0 V	5.0 V		OUT	"	E1 to EO	"	40	"
10 T_C = +25°C	t_{PLH1}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																			2	30	"
	t_{PHL1}																				"	39	"
	t_{PLH2}																				"	53	"
	t_{PHL2}																				"	44	"
	t_{PLH3}																				"	30	"
	t_{PHL3}																				"	58	"
	t_{PLH4}																				"	78	"
	t_{PHL4}																				"	34	"
	t_{PLH5}																				"	39	"
	t_{PHL5}																				"	39	"
	t_{PLH6}																				"	29	"
	t_{PHL6}																				"	53	"
	t_{PLH7}																				"	34	"
	t_{PHL7}																				"	52	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Case X pins not referenced are NC.

2/ Input voltages shown are: A = 2.0 V minimum and B = 0.7 V maximum.

3/ Output voltages shall be either:

a. H = 2.5 V minimum and L = 0.4 V maximum when using a high-speed checker double comparator; or
b. H ≥1.5 V and L ≤1.5 V when using a high-speed checker single comparator.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X 1/ / Test no.	1 / 2 / 4	2 / 3 / 5	3 / 4 / 6	4 / 5 / 7	5 / 7 / E1	6 / 8 / A2	7 / 9 / A1	8 / 10 / GND	9 / 12 / A0	10 / 13 / 0	11 / 14 / 1	12 / 15 / 2	13 / 17 / 3	14 / 18 / GS	15 / 19 / E0	16 / 20 / V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5	2.0 V "	2.0 V "	2.0 V "	2.0 V "	0.7 V " 2.0 V 2.0 V	-400 µA	-400 µA	GND " " " "	-400 µA	0.7 V "	2.0 V "	2.0 V "	2.0 V "	-400 µA	-400 µA	4.5 V " " " "	A2 A1 A0 GS E0	2.5 " " " "		V " " " "
	V_{OL}	3007	6 7 8 9 10	2.0 V	2.0 V	2.0 V	0.7 V " " " 2.0 V	0.7 V " " " "	12 mA	12 mA	" " " " "	12 mA	2.0 V	2.0 V	2.0 V	2.0 V	4 mA	4 mA	" " " " "	A2 A1 A0 GS E0		0.4 " " " "	" " " " "
	I_{IL1}	3009	11 12 13 14	0.4 V 5.5 V "	5.5 V 0.4 V 5.5 V	5.5 V 5.5 V 0.4 V 5.5 V	5.5 V " " 0.4 V	GND " " "			" " " "		5.5 V " " "	5.5 V " " "	5.5 V " " "	5.5 V " " "			5.5 V " " "	4 5 6 7	-.26	-.76 " " "	mA " " "
	I_{IL2}	"	15	"	"	"	5.5 V	"			"		0.4 V	"	"	"			"	0	-.16	-.40	"
	I_{IL1}	" " "	16 17 18	" " "	" " "	" " "	" " "	" " "			" " "		5.5 V "	0.4 V 5.5 V 5.5 V	" 0.4 V 5.5 V	" " 0.4 V			" " "	1 2 3	-.26 " "	-.76 " "	" " "
	I_{IL2}	"	19					0.4 V			"								"	E1	-.12	-.36	"
	I_{IH1}	3010 " " " " " " "	20 21 22 23 24 25 26 27	2.7 V "	2.7 V "	2.7 V "	2.7 V "				" " " " " " " "		2.7 V "	2.7 V "	2.7 V "	2.7 V "			" " " " " " " "	4 5 6 7 0 1 2 3		40 " " " " " " "	µA " " " " " " "
	I_{IH2}	"	28					2.7 V			"								"	E1		70	"
	I_{IH3}	" " " " " " "	29 30 31 32 33 34 35 36	5.5 V "	5.5 V "	5.5 V "	5.5 V "				" " " " " " " "		5.5 V "	5.5 V "	5.5 V "	5.5 V "			" " " " " " " "	4 5 6 7 0 1 2 3		200 " " " " " " "	" " " " " " " "
	I_{IH4}	"	37					5.5 V			"								"	E1		100	"
	I_{OS}	3011	38 39 40 41 42	4.5 V "	4.5 V "	4.5 V "	4.5 V "	GND " 4.5 V 4.5 V	GND	GND	" " " " "	GND	GND "	4.5 V "	4.5 V "	4.5 V "	GND	GND	" " " " "	A2 A1 A0 GS E0	-30 " " -15 -15	-130 " " -100 -100	mA " " " "

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Cases X 1/ Test no.	1 2 4	2 3 5	3 4 6	4 5 7	5 7 E1	6 8 A2	7 9 A1	8 10 GND	9 12 A0	10 13 0	11 14 1	12 15 2	13 17 3	14 18 GS	15 19 E0	16 20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{CC1}	3005	43				GND	GND			GND								5.5 V	V_{CC}		25	mA
	I_{CC2}	3005	44								"								"	V_{CC}		23	"
	I_{OZH}		45	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V			"	2.7 V	2.0 V	2.0 V	2.0 V	2.0 V			"	A0		20	uA
			46	"	"	"	"	"	2.7 V	2.7 V	"		"	"	"	"			"	A1		"	"
			47	"	"	"	"	"			"		"	"	"	"			"	A2		"	"
	I_{OZL}		48	"	"	"	"	"			"	0.4 V	"	"	"	"			"	A0		-20	"
			49	"	"	"	"	"		0.4 V	"		"	"	"	"			"	A1		"	"
			50	"	"	"	"	"	0.4 V		"		"	"	"	"			"	A2		"	"
	V_{IC}		51	-18 mA							"								4.5 V	4		-1.5	V
			52		-18 mA						"								"	5		"	"
			53			-18 mA					"								"	6		"	"
			54				-18 mA				"								"	7		"	"
			55					-18 mA			"								"	E1		"	"
			56								"		-18 mA						"	0		"	"
			57								"			-18 mA					"	1		"	"
			58								"				-18 mA				"	2		"	"
			59								"								"	3		"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 2/ T_C = +25°C	Truth table tests	3014	60	A	A	A	A	A	A	H	H	GND	H	A	A	A	A	H	H	5.0 V	3/ 4/			
		"	61	"	"	"	"	A	B	H	H	"	H	"	"	"	"	L	L	"				
		"	62	"	"	"	"	B	"	L	L	"	L	"	"	"	"	H	"	"				
		"	63	"	"	B	B	A	"	L	H	"	H	"	"	"	"	L	"	"				
		"	64	"	B	A	"	"	"	H	L	"	L	"	"	"	"	H	"	"				
		"	65	B	A	"	"	"	"	H	H	"	H	"	"	"	B	"	"	"				
		"	66	A	"	"	"	"	H	L	L	"	L	"	"	B	A	"	"	"				
		"	67	"	"	"	"	"	"	L	H	"	H	"	B	A	"	"	"	"				
		"	68	"	"	"	"	"	"	H	L	"	L	B	A	"	"	"	"	"				
		"	69	"	"	"	"	"	"	"	H	"	H	A	"	"	"	"	"	"				
		"	70	"	"	"	"	"	"	"	H	"	H	B	"	"	"	"	"	"				
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 Fig. 5	71	5.0 V	5.0 V	5.0 V	IN	GND		OUT	GND	OUT		5.0 V	5.0 V	5.0 V			5.0 V	7 to A0	2	22	ns
		"	72	"	"	"	"	"	OUT	"	"			"	"	"			"	7 to A1	"	"	"
		"	73	"	"	"	5.0 V	"		OUT	"			"	"	"			"	7 to A2	"	"	"
		"	74	"	"	IN	"	"	OUT	"	"	OUT		"	"	"			"	6 yo A1	"	"	"
		"	75	"	"	IN	"	"		"	"			"	"	"			"	6 to A2	"	"	"
		"	76	"	IN	5.0 V	"	"	OUT	"	"		OUT	"	"	"			"	5 to A0	"	"	"
		"	77	"	IN	"	"	"	OUT	"	"			"	"	"			"	5 to A2	"	"	"
		"	78	IN	5.0 V	"	"	"		"	"		OUT	"	"	"			"	4 to A2	"	"	"
		"	79	5.0 V	"	"	"	"		"	"			"	"	IN			"	3 to A0	"	"	"
		"	80	"	"	"	"	"		OUT	"			"	"	IN			"	3 to A1	"	"	"
		"	81	"	"	"	"	"		OUT	"			"	IN	5.0 V			"	2 to A1	"	"	"
		"	82	"	"	"	"	"			"		OUT	IN	5.0 V	5.0 V			"	1 to A0	"	"	"

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X 1/ / Test no.	1 / 2 / 4	2 / 3 / 5	3 / 4 / 6	4 / 5 / 7	5 / 7 / E1	6 / 8 / A2	7 / 9 / A1	8 / 10 / GND	9 / 12 / A0	10 / 13 / 0	11 / 14 / 1	12 / 15 / 2	13 / 17 / 3	14 / 18 / GS	15 / 19 / E0	16 / 20 / V_{CC}	Measured terminal	Test Limits Min	Test Limits Max	Unit
9 T_C = +25°C	t_{PHL1}	3003 Fig. 5	83	5.0 V	5.0 V	5.0 V	IN	GND			GND	OUT		5.0 V	5.0 V	5.0 V			5.0 V	7 to A0	2	35	ns
			84	"	"	"	"	"		OUT	"			"	"	"			"	7 to A1	"	"	"
			85	"	"	"	"	"	OUT		"			"	"	"			"	7 to A2	"	"	"
			86	"	"	IN	5.0 V	"		OUT	"			"	"	"			"	6 to A1	"	"	"
			87	"	"	IN	"	"	OUT		"			"	"	"			"	6 to A2	"	"	"
			88	"	IN	5.0 V	"	"			"	OUT		"	"	"			"	5 to A0	"	"	"
			89	"	IN	"	"	"	OUT		"			"	"	"			"	5 to A2	"	"	"
			90	IN	5.0 V	"	"	"	OUT		"			"	"	"			"	4 to A2	"	"	"
			91	5.0 V	"	"	"	"			"	OUT		"	"	IN			"	3 to A0	"	"	"
			92	"	"	"	"	"		OUT	"			"	"	IN			"	3 to A1	"	"	"
			93	"	"	"	"	"		OUT	"			"	IN	5.0 V			"	2 to A1	"	"	"
			94	"	"	"	"	"			"	OUT		IN	5.0 V	5.0 V			"	1 to A0	"	"	"
	t_{PLH2}	"	95	IN	"	"	"	"		OUT	"					GND			"	4 to A1	"	40	"
		"	96	IN	"	"	"	"			"	OUT				GND			"	4 to A0	"	"	"
		"	97		GND	IN	"	"			"	OUT							"	6 to A0	"	"	"
	t_{PHL2}	"	98	IN	5.0 V	5.0 V	"	"		OUT	"					GND			"	4 to A1	"	"	"
		"	99	IN	5.0 V	5.0 V	"	"			"	OUT				GND			"	4 to A0	"	"	"
		"	100		GND	IN	"	"			"	OUT							"	6 to A0	"	"	"
	t_{PLH3}	"	101	5.0 V	5.0 V	5.0 V	"	"			"		IN	5.0 V	5.0 V	5.0 V		OUT	"	0 to E0	"	23	"
		"	102	"	"	"	"	"			"		5.0 V	IN	5.0 V	"		"	"	1 to E0	"	"	"
		"	103	"	"	"	"	"			"		"	5.0 V	IN	"		"	"	2 to E0	"	"	"
		"	104	"	"	"	"	"			"		"	"	5.0 V	IN		"	"	3 to E0	"	"	"
		"	105	IN	"	"	"	"			"		"	"	"	5.0 V		"	"	4 to E0	"	"	"
		"	106	5.0 V	IN	"	"	"			"		"	"	"	"		"	"	5 to E0	"	"	"
		"	107	"	5.0 V	IN	"	"			"		"	"	"	"		"	"	6 to E0	"	"	"
		"	108	"	"	5.0 V	IN	"			"		"	"	"	"		"	"	7 to E0	"	"	"
	t_{PHL3}	"	109	"	"	"	5.0 V	"			"		IN	"	"	"		"	"	0 to E0	"	45	"
		"	110	"	"	"	"	"			"		5.0 V	IN	"	"		"	"	1 to E0	"	"	"
		"	111	"	"	"	"	"			"		"	5.0 V	IN	"		"	"	2 to E0	"	"	"
		"	112	"	"	"	"	"			"		"	"	5.0 V	IN		"	"	3 to E0	"	"	"
		"	113	IN	"	"	"	"			"		"	"	"	5.0 V		"	"	4 to E0	"	"	"
		"	114	5.0 V	IN	"	"	"			"		"	"	"	"		"	"	5 to E0	"	"	"
		"	115	"	5.0 V	IN	"	"			"		"	"	"	"		"	"	6 to E0	"	"	"
		"	116	"	"	5.0 V	IN	"			"		"	"	"	"		"	"	7 to E0	"	"	"
	t_{PLH4}	"	117	"	"	"	5.0 V	"			"		IN	"	"	"	OUT	"	"	0 to GS	"	60	"
		"	118	"	"	"	"	"			"		5.0 V	IN	"	"	"	"	"	1 to GS	"	"	"
		"	119	"	"	"	"	"			"		"	5.0 V	IN	"	"	"	"	2 to GS	"	"	"
		"	120	"	"	"	"	"			"		"	"	5.0 V	IN	"	"	"	3 to GS	"	"	"
		"	121	IN	"	"	"	"			"		"	"	"	5.0 V	"	"	"	4 to GS	"	"	"
		"	122	5.0 V	IN	"	"	"			"		"	"	"	"	"	"	"	5 to GS	"	"	"
		"	123	"	5.0 V	IN	"	"			"		"	"	"	"	"	"	"	6 to GS	"	"	"
		"	124	"	"	5.0 V	IN	"			"		"	"	"	"	"	"	"	7 to GS	"	"	"
	t_{PHL4}	"	125	"	"	"	5.0 V	"			"		IN	"	"	"	"	"	"	0 to GS	"	26	"
		"	126	"	"	"	"	"			"		5.0 V	IN	"	"	"	"	"	1 to GS	"	"	"
		"	127	"	"	"	"	"			"		"	5.0 V	IN	"	"	"	"	2 to GS	"	"	"
		"	128	"	"	"	"	"			"		"	"	5.0 V	IN	"	"	"	3 to GS	"	"	"
		"	129	IN	"	"	"	"			"		"	"	"	5.0 V	"	"	"	4 to GS	"	"	"
		"	130	5.0 V	IN	"	"	"			"		"	"	"	"	"	"	"	5 to GS	"	"	"
		"	131	"	5.0 V	IN	"	"			"		"	"	"	"	"	"	"	6 to GS	"	"	"
		"	132	"	5.0 V	5.0 V	IN	"			"		"	"	"	"	"	"	"	7 to GS	"	"	"

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Cases X 1/ Test no.	1 / 2 / 4	2 / 3 / 5	3 / 4 / 6	4 / 5 / 7	5 / 7 / E1	6 / 8 / A2	7 / 9 / A1	8 / 10 / GND	9 / 12 / A0	10 / 13 / 0	11 / 14 / 1	12 / 15 / 2	13 / 17 / 3	14 / 18 / GS	15 / 19 / E0	16 / 20 / V_{CC}	Measured terminal	Test limits: Min	Max	Unit
9 T_C = +25°C	t_{PLH6}	3003 Fig. 5	133				GND	IN			GND						OUT		5.0 V	E1 to GS		22	ns
	t_{PHL6}	"	134				GND	"			"						"		"	E1 to GS		41	"
	t_{PLH7}	"	135	5.0 V	5.0 V	5.0 V	5.0 V	"			"		5.0 V	5.0 V	5.0 V	5.0 V		OUT	"	E1 to E0	"	26	"
	t_{PHL7}	"	136	5.0 V	5.0 V	5.0 V	5.0 V	"			"		5.0 V	5.0 V	5.0 V	5.0 V		OUT	"	E1 to E0	"	40	"
	t_{PZL}	"	137				GND	"			"	OUT							"	E1 to A0		46	"
		"	138				"	"		OUT	"								"	E1 to A1		"	"
		"	139				"	"	OUT		"								"	E1 to A2		"	"
	t_{PLZ}	"	140				"	"			"	OUT							"	E1 to A0		40	"
		"	141				"	"		OUT	"								"	E1 to A1		"	"
		"	142				"	"	OUT		"								"	E1 to A2		"	"
	t_{PZH}	"	143	5.0 V	5.0 V	5.0 V	5.0 V	"			"	OUT	GND	5.0 V	5.0 V	5.0 V			"	E1 to A0		49	"
		"	144	"	"	"	"	"		OUT	"		"	"	"	"			"	E1 to A1		"	"
		"	145	"	"	"	"	"	OUT		"		"	"	"	"			"	E1 to A2		"	"
	t_{PHZ}	"	146	"	"	"	"	"			"	OUT	"	"	"	"			"	E1 to A0		32	"
		"	147	"	"	"	"	"		OUT	"		"	"	"	"			"	E1 to A1		"	"
		"	148	"	"	"	"	"	OUT		"		"	"	"	"			"	E1 to A2		"	"
10 T_C = +125°C	t_{PLH1}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																			2	29	"
	t_{PHL1}																				"	46	"
	t_{PLH2}																				"	52	"
	t_{PHL2}																				"	52	"
	t_{PLH3}																				"	30	"
	t_{PHL3}																				"	58	"
	t_{PLH4}																				"	78	"
	t_{PHL4}																				"	34	"
	t_{PLH6}																				"	29	"
	t_{PHL6}																				"	53	"
	t_{PLH7}																				"	34	"
	t_{PHL7}																				"	52	"

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits		Unit
			Cases X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	4	5	6	7	E1	A2	A1	GND	A0	0	1	2	3	GS	E0	V_{CC}				
10 T_C = +25°C	t_{PZL}																				2	60	ns
	t_{PLZ}																				"	52	"
	t_{PZH}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																			"	64	"
	t_{PHZ}																				"	42	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Case X pins not referenced are NC.

2/ Input voltages shown are: A = 2.0 V minimum and B = 0.7 V maximum.

3/ Output voltages shall be either:

a. H = 2.5 V minimum and L = 0.4 V maximum when using a high-speed checker double comparator; or
b. H ≥1.5 V and L ≤1.5 V when using a high-speed checker single comparator.

4/ Apply resistor 0.5 kΩ to 5.0 kΩ between pin 16 and pins 6, 7, and 9.

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II, herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical paramters shall be as specified in table II, herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical test parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

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e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable, these requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
I_{IN}	Current flowing into an input terminal.
V_{IN}	Voltage level at an input level.
t_{PHZ}	Output disable time (of a three-state output) from high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high level to a high impedance (off) state.
t_{PLZ}	Output disable time (of a three-state output) from low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined low level to a high-impedance (off) state.
t_{PZH}	Output enable time (of a three-state output) to high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high impedance (off) state to the defined high level.
t_{PZL}	Output enable time (of a three-state output) to low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high impedance (off) state to the defined low level.
I_{OZH}	Output current in the high-impedance mode with the output voltage high.
I_{OZL}	Output current in the high-impedance mode with the output voltage low.

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS148
02	54LS348

6.7 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit A
	Texas Instruments
01	X
02	X

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/360A	MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY TTL, PRIORITY ENCODE

3a. NAME OF SUBMITTING ORGANIZATION MONOLITHIC SILICON	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
8a. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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